Multivariable Taylor's Theorem

Background Notes for Project 8

Draft prepared for further editing

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Abstract

These notes collect the main definitions and theorem statements needed for Project 8 on multivariable Taylor's theorem. The central source is Edwards' *Advanced Calculus of Several Variables*, especially Chapter II, Sections 2, 3, 6, and 7. The goal is to make the theorem self-contained enough to support a formal project paper and a short presentation.

1 Directional Derivatives

Definition 1.1 (Directional derivative)**.** *Let $U \subset \mathbb{R}^n$ be open, let $f : U \to \mathbb{R}$, let $a \in U$, and let $h \in \mathbb{R}^n$. The directional derivative of f at a in the direction h, when it exists, is*

$$D_h f(a) = \lim_{t \to 0} \frac{f(a + th) - f(a)}{t}.$$

Proposition 1.2 (Gradient formula for directional derivatives)**.** *If $f : U \to \mathbb{R}$ is differentiable at a, then the directional derivative exists for every $h \in \mathbb{R}^n$, and*

$$D_h f(a) = df_a(h) = \nabla f(a) \cdot h = \sum_{i=1}^{n} h_i D_i f(a).$$

This proposition is the bridge between the geometric meaning of a directional derivative and the algebraic notation used in Taylor's formula. In particular, when f is sufficiently smooth, the operator D_h may be written as

$$D_h = h_1 D_1 + \cdots + h_n D_n.$$

2 Iterated Directional Derivatives

Definition 2.1 (Class C^k)**.** *Let $U \subset \mathbb{R}^n$ be open. A real-valued function $f : U \to \mathbb{R}$ is of class C^k on U if all iterated partial derivatives of order at most k exist and are continuous on U.*

Definition 2.2 (Iterated directional derivative)**.** *Let $f \in C^k(U)$ and $h \in \mathbb{R}^n$. Define*

$$D_h^0 f = f, \qquad D_h^k f = D_h(D_h^{k-1} f) \quad (k \geq 1).$$

Thus $D_h^k f$ means that the directional derivative in the same direction h has been applied k times.

Proposition 2.3 (Multinomial form)**.** *Let $f \in C^k(U)$. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n)$, write*

$$|\alpha| = \alpha_1 + \cdots + \alpha_n, \qquad h^\alpha = h_1^{\alpha_1} \cdots h_n^{\alpha_n}, \qquad D^\alpha f = D_1^{\alpha_1} \cdots D_n^{\alpha_n} f.$$

Then

$$D_h^k f(x) = \sum_{|\alpha|=k} \frac{k!}{\alpha_1! \cdots \alpha_n!} h^\alpha D^\alpha f(x).$$

Proof. Since $D_h = h_1 D_1 + \cdots + h_n D_n$, the k-fold iterate is $(h_1 D_1 + \cdots + h_n D_n)^k$. The multinomial theorem expands this operator into the displayed sum. The hypothesis $f \in C^k(U)$ ensures that the iterated partial derivatives may be reordered without changing their value. □

3 Taylor Polynomial

Definition 3.1 (Taylor polynomial in the direction h)**.** *Let $f \in C^r(U)$, $a \in U$, and suppose that the relevant line segment near a lies in U. The rth Taylor polynomial of f at a, written as a polynomial in the increment h, is*

$$P_r(h) = \sum_{k=0}^{r} \frac{D_h^k f(a)}{k!}.$$

Using the multinomial form of D_h^k, this is equivalently

$$P_r(h) = \sum_{|\alpha| \le r} \frac{D^\alpha f(a)}{\alpha_1! \cdots \alpha_n!} h^\alpha.$$

4 Multivariable Taylor's Theorem

Theorem 4.1 (Multivariable Taylor theorem with Lagrange remainder)**.** *Let $U \subset \mathbb{R}^n$ be open, let $a \in U$, let $h \in \mathbb{R}^n$, and suppose that the line segment*

$$L = \{a + th : 0 \le t \le 1\}$$

is contained in U. If $f : U \to \mathbb{R}$ is of class C^{r+1}, then there is a point $c \in L$ such that

$$f(a+h) = \sum_{k=0}^{r} \frac{D_h^k f(a)}{k!} + \frac{D_h^{r+1} f(c)}{(r+1)!}.$$

Proof. Define a one-variable function $g : [0,1] \to \mathbb{R}$ by

$$g(t) = f(a + th).$$

By the one-variable Taylor theorem, there is a $\theta \in (0,1)$ such that

$$g(1) = \sum_{k=0}^{r} \frac{g^{(k)}(0)}{k!} + \frac{g^{(r+1)}(\theta)}{(r+1)!}.$$

It remains to identify the derivatives of g. By the chain rule,

$$g'(t) = df_{a+th}(h) = D_h f(a + th).$$

Inductively, if $g^{(k)}(t) = D_h^k f(a + th)$, then applying the same chain-rule argument to $D_h^k f$ gives

$$g^{(k+1)}(t) = D_h^{k+1} f(a + th).$$

Therefore $g^{(k)}(0) = D_h^k f(a)$, and

$$g^{(r+1)}(\theta) = D_h^{r+1} f(a + \theta h).$$

Set $c = a + \theta h$. Since $c \in L$, the desired formula follows. $\square$

Remark 4.2. *This is the version stated in Edwards as Taylor's formula in several variables. It is essentially the one-variable theorem applied to the restriction of f to the line segment from a to $a + h$.*

5 Example of Order Two

Example 5.1. *Let $f(x, y) = e^{x+2y}$, let $a = (0, 0)$, and write $h = (x, y)$. Since*

$$D_h f(u, v) = (x + 2y)e^{u+2v},$$

we have

$$D_h^k f(0, 0) = (x + 2y)^k.$$

The second-order Taylor formula gives

$$e^{x+2y} = 1 + (x + 2y) + \frac{(x + 2y)^2}{2} + \frac{(x + 2y)^3 e^{\theta(x+2y)}}{6}$$

for some $\theta \in (0, 1)$. This illustrates how a multivariable Taylor expansion can be obtained by applying the one-variable Taylor theorem along the line $t \mapsto t(x, y)$.

6 How This Fits Project 8

The project asks for three main tasks:

1. explain iterated directional derivatives;
2. prove the multivariable Taylor theorem;
3. give examples, including at least one multivariable example of order $r \geq 2$.

The definitions, propositions, theorem, proof, and example above form a first draft of that mathematical core. A full paper should add more narrative, additional examples, and a polished introduction that explains why polynomial approximations matter.

References

[1] C. H. Edwards, Jr., *Advanced Calculus of Several Variables*, Academic Press, 1973. See Chapter II, Sections 2, 3, 6, and 7, especially Theorem 7.1.

[2] MATH 5023-01 Multivariable Analysis, *Project 8: Multivariable Taylor's Theorem*, Spring 2026 project description.